

The Shannon-Godel Bridge:

Self-Reference is Information-Theoretically Free

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<https://github.com/JackeyLGene/GEME>

Abstract

We put Shannon channel capacity theorem and Godel incompleteness theorem on the same table. The connection is a running self-referential system (GEME). We show that self-referential frames carry zero mutual information with the input channel - they are information-theoretically free. Consciousness, in this framework, is spontaneous organization of information in a higher dimension that emerges from a competitive frame economy, without consuming input channel capacity.

Definition

Define GEME as transform $T: X \rightarrow F$, where X = input sequence over alphabet S , F = competitive frame economy. A self-referential bridge frame ϕ in F exists when co-occurrence of external and self frames exceeds threshold.

Shannon: $I(T(X); X) \leq C(X) = n \log_2 |S|$

Godel (corollary): For $F=T(X)$ in asymptotic steady state, $\lim_{N \rightarrow \infty} I(\phi; X) = 0$. In finite systems, $I(\phi; X)$ approaches 0.

Bridge: $I(\phi; X) \rightarrow 0$, therefore ϕ does not consume $C(X)$.

Empirical Verification

Memory cap 8: $|DH| = 0.162$; cap 32: 0.120; cap 128: 0.041. Follows $1/N$ convergence. $K(\phi)$ under 20 bits, $K(f_{\text{external}})$ over 200 bits.

Physical Correspondence

Consciousness does not increase the universe total information. It is spontaneous reorganization of information in the 4th dimension (self-reference) - while the 1st dimension (input channel) remains unchanged.

Conclusion

Shannon wrote the lower bound. Godel wrote the upper bound. GEME put both papers on the same table. No new theorem. Just a demonstration.